

Q1. You are analyzing a node in a sensor network where the sensors act as Current Sources. You may choose Nodal/Mesh Analysis. Justify your choice.

A. Initially, the resultant circuit is as shown in Figure.1 (a). Calculate I_0 flowing in $3\text{ k}\Omega$ resistor.

B. Then, the two sensors are connected as shown in Figure.1(b). Find V_1 across the $3\text{ k}\Omega$ resistor

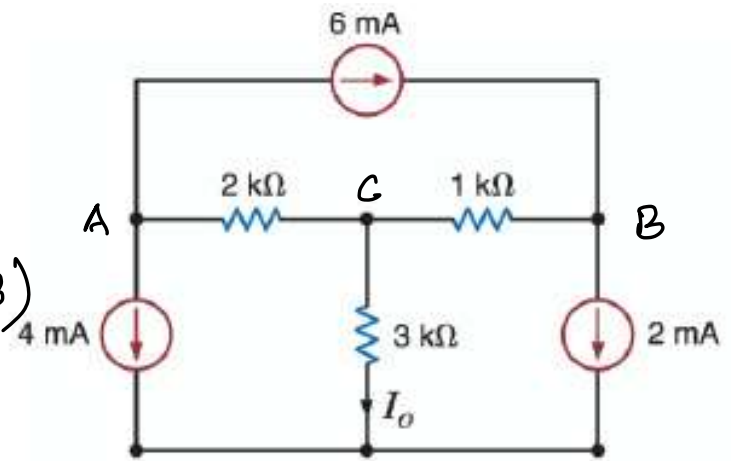
Steps

① find current thro
 $1\text{ k}\Omega$ (KCL - node B)

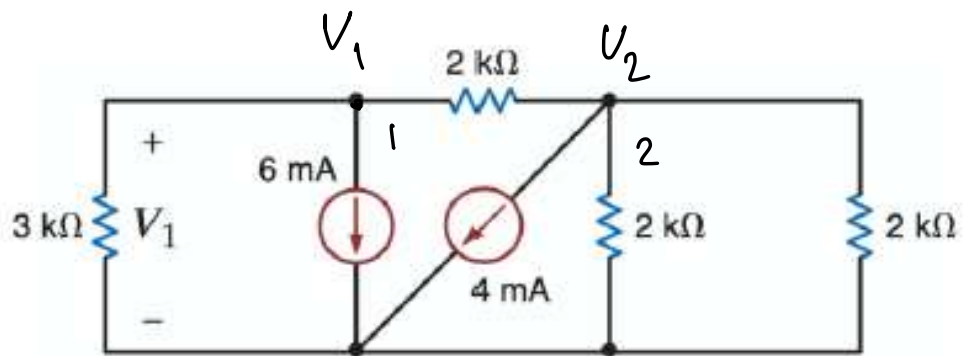
② Find current thro
 $2\text{ k}\Omega$ (KCL - node A)

③ find current thro $3\text{ k}\Omega$ (KCL - node C)
 $= I_0$

④ $V_1 = I_0 \times 3000$



B) Using nodal analysis



① Apply KCL to node 1

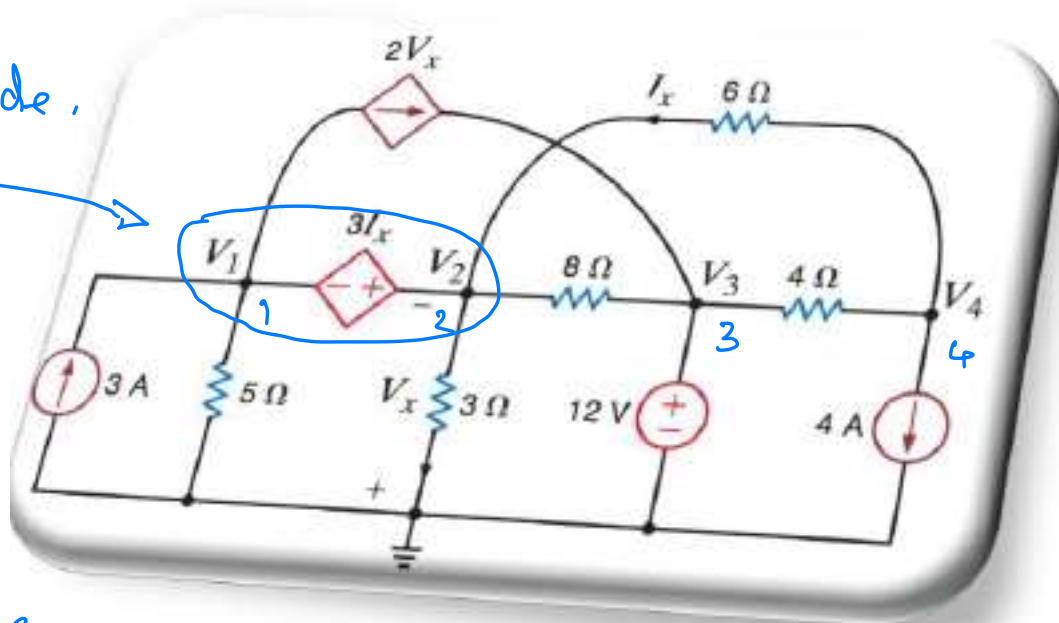
$$\frac{V_1}{3000} + 6 \times 10^{-3} + \frac{(V_1 - V_2)}{2000} = 0$$

② Apply KCL to node - 2

$$\frac{V_2 - V_1}{2000} + 4 \times 10^{-3} + \frac{V_2}{2000} + \frac{V_2}{2000} = 0$$

2. In many amplifier and sensor circuits, a voltage source establishes a fixed potential difference between two points, neither of which is ground. This creates a "floating" source that could complicate standard nodal analysis because the current passing through the voltage source is initially unknown. The given circuit consists of current dependent voltage source. Determine the node voltages in the given circuit

Supernode.



Note

$$V_x = -V_2$$

$$I_x = \frac{V_4 - V_2}{6}$$

At Supernode

$$-3 + \frac{V_1}{5} - 2V_2 + \frac{V_2 - V_4}{6} + \frac{V_1 - V_3}{8} + \frac{V_2}{3} = 0 \quad (1)$$

(currents going away from the node)

Within Supernode

$$V_1 + 3\left(\frac{V_4 - V_2}{6}\right) = V_2 \quad (2)$$

At node 3

$$V_3 = 12 \text{ V} \quad (3)$$

At node 4

$$\frac{V_4 - V_3}{4} + \frac{V_4 - V_2}{6} + 4 = 0 \quad (4)$$

Can be solved for V_1 to V_4 ,

3. You are analysing a representative power distribution circuit driven by independent and dependent voltage sources.

A. Find the power dissipated in the $6\text{ k}\Omega$ resistor for circuit shown in A.

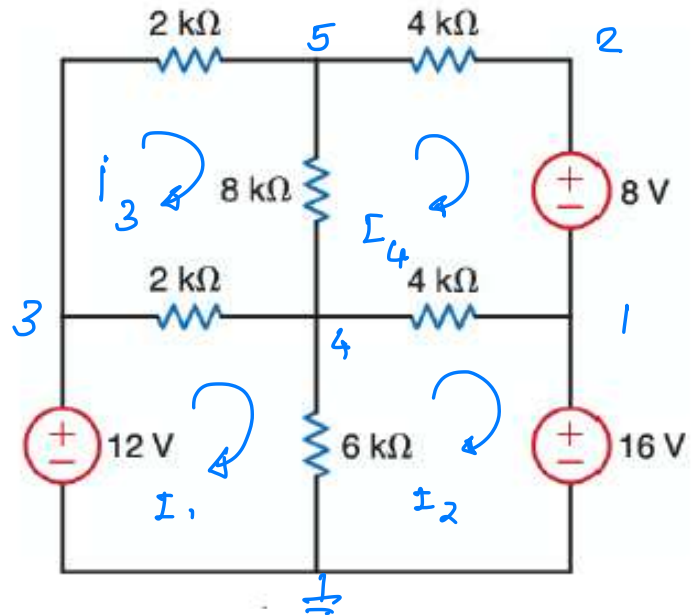
B. Find I_1 , I_2 , I_3 and I_4 for the circuit shown in B.

KVL

Mesh - 1

$$12 = 2(I_1 - I_3) + 6(I_1 - I_2)$$

- ①



Mesh - ②

$$16 + 6(I_2 - I_1) + 4(I_2 - I_4) = 0$$

Mesh - ③

$$2I_3 + 8(I_3 - I_4) + 2(I_3 - I_1) = 0$$

Mesh - ④

$$8 + 4(I_4 - I_2) + 8(I_4 - I_3) + 4I_4 = 0$$

★ Node analysis is simpler

as $V_1 = 16\text{ V}$; $V_2 = 24\text{ V}$, $V_3 = 12\text{ V}$

form node eqn for V_3 and V_4
and solve only two variables

3B Mesh analysis

Note $V_x = -5 I_1$
 $I_x = I_4 - I_3$

kvl / mesh - I

$$\begin{aligned} (5+2) I_1 + 4 (I_1 - I_2) \\ + 3 (I_4 - I_3) \\ + 7 (I_1 - I_3) = 0 \end{aligned}$$

- ①

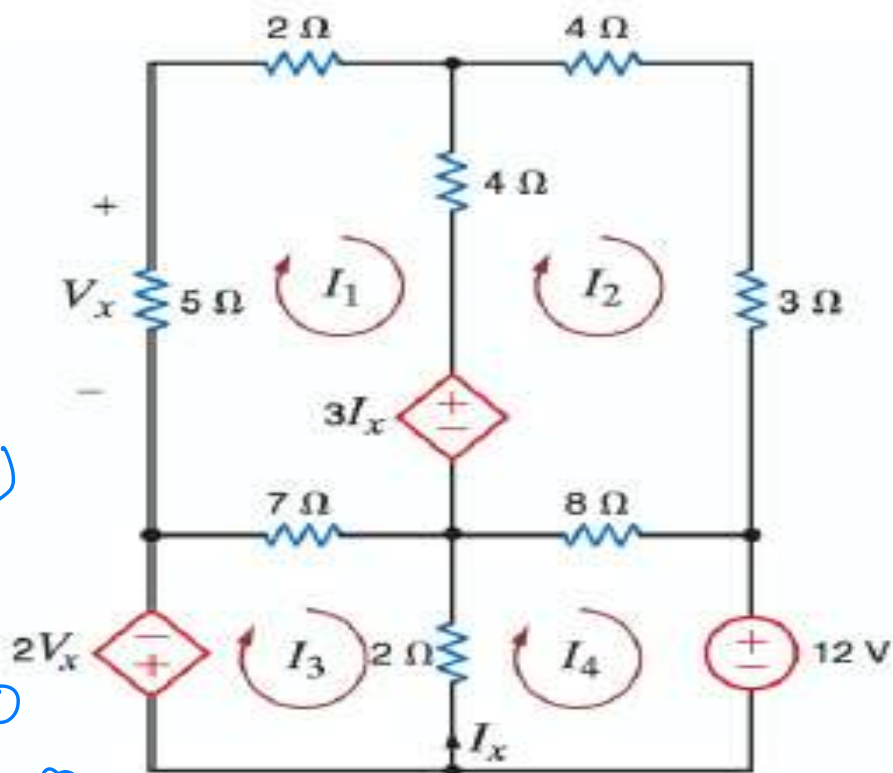


Figure B

Mesh-2

$$7 I_2 + 8 (I_1 - I_4) - 3 (I_4 - I_3) + 4 (I_2 - I_1) = 0$$

Mesh-3

$$7 (I_2 - I_1) + 2 (I_3 - I_4) + 2 (-5 I_1) = 0$$

Mesh-4

$$8 (I_4 - I_2) + 12 + 2 (I_4 - I_3) = 0$$

4. Some precision power supplies use a current regulator shared between two control loops. When a current source sits on the boundary between two meshes, you cannot write a standard KVL loop equation because the voltage drop across an ideal current source is undefined (it adjusts to whatever the circuit needs).

For the given circuit, find the power dissipated in the $3\ \Omega$ resistor.

Considering the current source
 5 A , a Super-mesh is
 considered as shown
 within super mesh

$$I_2 - I_1 = 5\text{ A} \quad \text{--- (1)}$$

In Super mesh

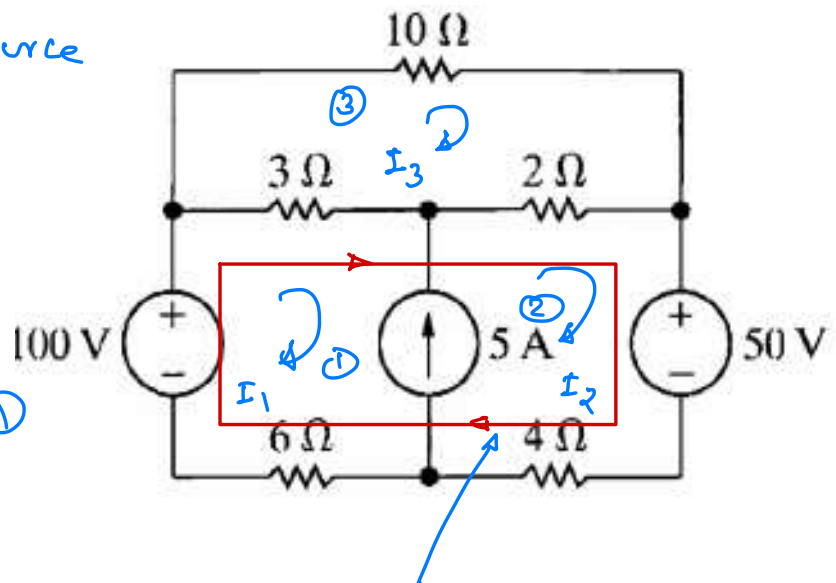
$$100 = 3(I_1 - I_3) + 2(I_2 - I_3) + 50 + 4I_2 + 6I_1 = 0 \quad \text{--- (2)}$$

In Mesh - 3

$$(I_3 - I_1)3 + 10I_3 + 2(I_3 - I_2) = 0 \quad \text{--- (3)}$$

I_1 & I_2 and I_3 can be solved now.

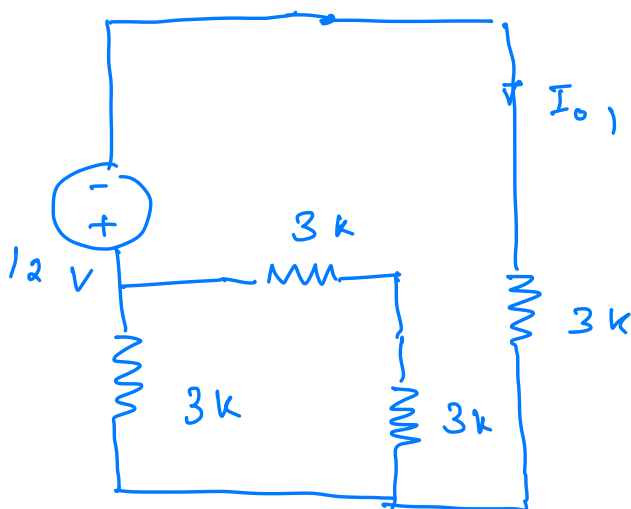
Power dissipated in $3\ \Omega$ resistor = $(I_3 - I_1)^2 \cdot 3$



Q5. You could encounter a circuit mixing the two different sources (e.g. a music signal modelled as a voltage source and a microphone signal modelled as a current source). Calculate the combined output voltage V_o for two given configurations using the superposition theorem.

There are two sources
 → Superposition principle
 can be applied

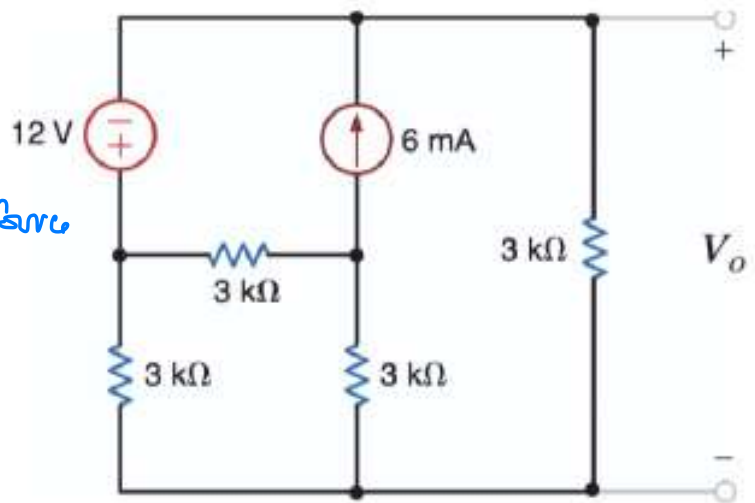
Ckt with only 12 V source



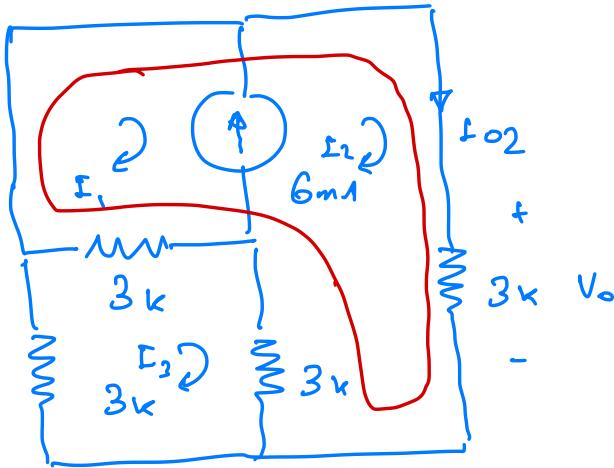
⇒

I_{o1} can be solved for
 from this ckt

$$V_{o1} = I_{o1} \times 3k$$



Circuit with only 6 mA source
Using mesh equation.



$$I_2 - I_1 = 6 \text{ mA}$$

$$3(I_2) + 3(I_3 - I_2) + 3(I_3 - I_1) = 0$$

$$3(I_1 - I_2) + 3I_2 + 3(I_2 - I_3) = 0$$

Solve for I_1, I_2, I_3

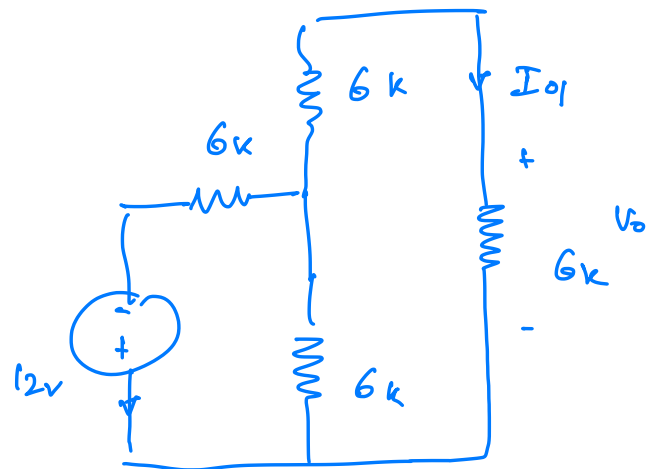
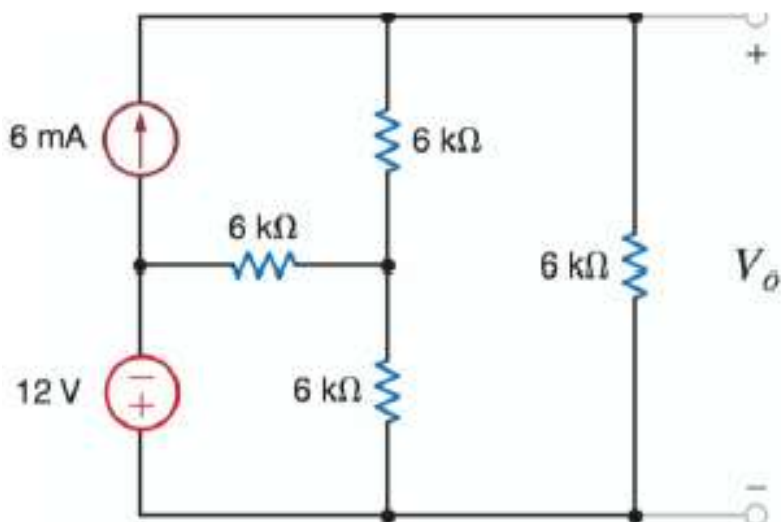
$$V_{02} = I_{02} \cdot 3$$

(Across 3k)
as shown $V_0 = V_{01} + V_{02}$

5 b

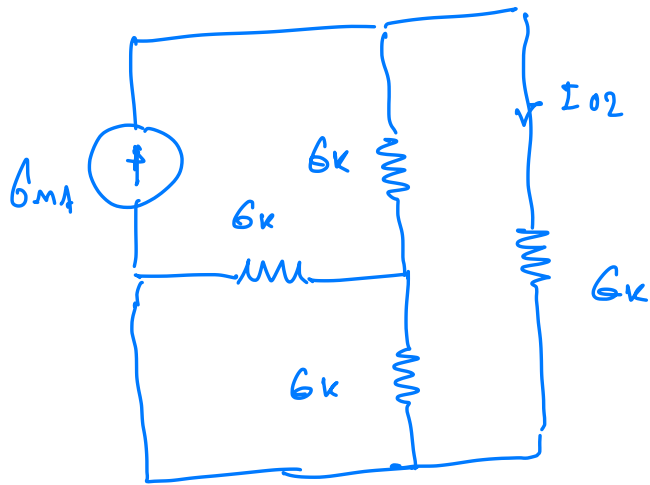
Apply Super position theorem

Ckt with only with 12 V source

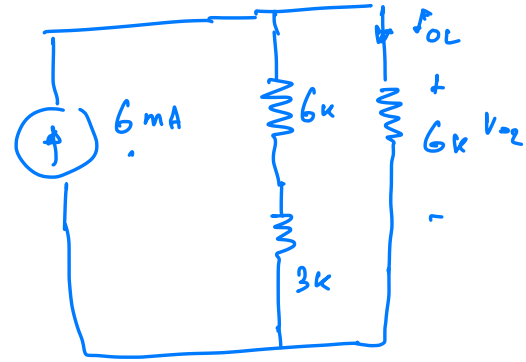


$$V_{01} = I_{01} \times 6000$$

Ckt with only 6 mA



$\Rightarrow$



$$V_{O2} = I_{O2} \times 6k$$